

Dot Product of Vectors

Algebra & Vectors Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Compute the dot product of two 2D vectors using the component formula
- Apply scalar multiplication before or after finding a dot product
- Identify and apply properties of the dot product including the commutative property

Problems

1. Find the dot product of the two vectors below:

$$\vec{u} = \langle 3, 4 \rangle, \quad \vec{v} = \langle 1, 2 \rangle$$

2. Find the dot product of the two vectors below:

$$\vec{u} = \langle 5, 2 \rangle, \quad \vec{v} = \langle 3, 6 \rangle$$

3. Find the dot product of the two vectors below:

$$\vec{u} = \langle -2, 5 \rangle, \quad \vec{v} = \langle 4, -3 \rangle$$

4. Use the commutative property to verify that the two dot products below are equal. Find both values:

$$\vec{u} = \langle 6, -1 \rangle, \quad \vec{v} = \langle 2, 4 \rangle$$

5. Find the dot product of any vector with the zero vector and state what you notice:

$$\vec{u} = \langle 7, -3 \rangle, \quad \vec{0} = \langle 0, 0 \rangle$$

6. First compute the scalar multiplication, then find the dot product:



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Labanan ang agwat.

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Vectors — Dot Product Worksheet

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$$\vec{u} = \langle 3, -1 \rangle, \quad 3\vec{v} \text{ where } \vec{v} = \langle 2, 5 \rangle$$

7. Find the dot product of vectors u and v , then multiply the result by vector w :

$$\vec{u} = \langle 2, 3 \rangle, \quad \vec{v} = \langle 1, -2 \rangle, \quad \vec{w} = \langle 4, -1 \rangle$$

8. Find the dot product of vector u with itself:

$$\vec{u} = \langle 3, 4 \rangle$$

9. Given the vectors below, compute the expression shown:

10. Determine the value of k so that the two vectors are perpendicular. Recall that two vectors are perpendicular when their dot product equals zero:

$$\vec{u} = \langle k, 3 \rangle, \quad \vec{v} = \langle 4, -8 \rangle$$



Dot Product of Vectors — Answer Key

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Answer Key

1. Answer: 11

- Multiply corresponding components: $3 \times 1 = 3$ and $4 \times 2 = 8$
- Add the products: $3 + 8 = 11$

2. Answer: 27

- Multiply corresponding components: $5 \times 3 = 15$ and $2 \times 6 = 12$
- Add the products: $15 + 12 = 27$

3. Answer: -23

- Multiply corresponding components: $-2 \times 4 = -8$ and $5 \times (-3) = -15$
- Add the products: $-8 + (-15) = -23$

4. Answer: Both equal 8

- Compute $u \cdot v$: $6 \times 2 + (-1) \times 4 = 12 - 4 = 8$
- Compute $v \cdot u$: $2 \times 6 + 4 \times (-1) = 12 - 4 = 8$
- Both results equal 8, confirming the commutative property

5. Answer: 0

- Multiply corresponding components: $7 \times 0 = 0$ and $-3 \times 0 = 0$
- Add the products: $0 + 0 = 0$
- Any vector dotted with the zero vector always equals 0

6. Answer: 3

- Compute $3v$: $3 \times \blacksquare 2, 5\blacksquare = \blacksquare 6, 15\blacksquare$
- Compute $u \cdot 3v$: $3 \times 6 + (-1) \times 15 = 18 - 15 = 3$

7. Answer: $\blacksquare -16, 4\blacksquare$

- Compute $u \cdot v$: $2 \times 1 + 3 \times (-2) = 2 - 6 = -4$
- Multiply scalar -4 by vector w : $-4 \times \blacksquare 4, -1\blacksquare = \blacksquare -16, 4\blacksquare$

8. Answer: 25

- Compute $u \cdot u$: $3 \times 3 + 4 \times 4 = 9 + 16 = 25$
- Note: $u \cdot u$ equals the square of the magnitude of u , since $|u| = 5$ and $5^2 = 25$

9. Answer: $\blacksquare -6, 26\blacksquare$

- Compute $u \cdot v$: $(-1)(2) + (3)(-4) = -2 - 12 = -14$
- Multiply by w : $-14 \times \blacksquare 1, -2\blacksquare = \blacksquare -14, 28\blacksquare$
- Compute $u \cdot w$: $(-1)(1) + (3)(-2) = -1 - 6 = -7$
- Multiply by 2: $2 \times (-7) = -14$, so $2(u \cdot w) = -14$



- Add scalar -14 to each component of $\begin{bmatrix} -14 \\ 28 \end{bmatrix}$: $\begin{bmatrix} -14 + (-14) \\ 28 + (-14) \end{bmatrix}$... wait — scalar added to vector is invalid; recompute: $2(\mathbf{u} \cdot \mathbf{w}) \cdot \mathbf{w} = 2(-7) \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -14 \\ 28 \end{bmatrix}$; total = $\begin{bmatrix} -14 + (-14) \\ 28 + 28 \end{bmatrix} = \begin{bmatrix} -28 \\ 56 \end{bmatrix}$ if both multiplied by $\mathbf{w}$. Using $(\mathbf{u} \cdot \mathbf{v})\mathbf{w} + 2(\mathbf{u} \cdot \mathbf{w})\mathbf{w}$: $\begin{bmatrix} -14 \\ 28 \end{bmatrix} + \begin{bmatrix} -14 \\ 28 \end{bmatrix} = \begin{bmatrix} -28 \\ 56 \end{bmatrix}$

10. Answer: $k = 6$

- Set up the dot product equal to zero: $k \times 4 + 3 \times (-8) = 0$
- Simplify: $4k - 24 = 0$
- Solve for k : $4k = 24$, so $k = 6$

